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A review on the quality of wood from agroforestry systems

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Abstract Agroforestry systems (AFS) are an established and well-documented practice with widely recognized economic, social, and ecosystem benefits. However, literature regarding their woody component analysis for wood products is still incipient. This study aimed to survey articles that report results on the quality of wood produced in agroforestry systems, identify existing knowledge gaps regarding the quality of wood from AFS and, with those results, guide new studies. A search on Scopus and Web of Science was conducted using terms related to agroforestry systems and wood quality. The findings were screened and analyzed, and the main data and wood characteristics of each fitting article were described. This review describes thirteen articles, comprising four countries (Brazil, Costa Rica, France, India and Portugal), and it discusses the properties and potential use of wood from twelve species: *Castanea sativa*, *Cedrela odorata*, *Eucalyptus grandis* × *Eucalyptus urophylla*, *Juglans nigra* × *regia*, *Khaya senegalensis*, *Parapiptadenia rigida*, *Peltophorum dubium*, *Populus deltoides* × *Populus tristris*, *Quercus robur*, *Quercus rotundifolia*, *Schizolobium parahyba* and *Tectona grandis*. Most of the woods had superior or similar characteristics to monoculture ones, except for *Tectona grandis*. The geographic distribution of those studies is limited, with no studies from Africa and Oceania, and few authors publishing on this subject. Even though AFS is a traditional agricultural practice, few studies address the quality of the wood from this system. This study gathers existing information about the quality of wood produced in AFS, highlighting the knowledge gaps on this theme and indicating improvements for future work.

Keywords Agroforestry · Agroforestry-pastoral system · AFS · Integrated crop-livestock-forestry · Wood technology

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Introduction

Agroforestry systems (AFS, also called agroforestry and integrated systems) are definied as the combination tree and/or shrub species with agricultural and/or animal production, according to a spatial and temporal arrangement, with high heterogeneity of its components (Nair et al. 2021; Peneireiro et al. 2000). Worldwide, 1.6 billion ha of land has potential to be under agroforestry management, while, currently, those systems are majorly located in tropical and subtropical lands[1] (Nair and Garrity 2012; Nair 2014), overall presenting structural complex systems (Pinto 2020). In Europe, there is an estimated implanted area of about 20 million ha of AFS (Mosquero-Losada et al. 2012; AGFORWARD 2017), with a wide diversity of species as a result of different climatic and cultural aspects, with more prominent pratices in the Mediterranean and boreal regions (Báder et al. 2023).

AFS enhance ecological interactions (Steenbock et al. 2020; Nair et al. 2021); contribute to rural livelihoods and nutritional security (Feliciano et al. 2018; Thomas et al. 2021); to socioeconomic development in land-constrained countries (Mankhin et al. 2022); improve site occupancy and utilization of different vertical strata (Feliciano et al. 2018); contribute to carbon stock/sequestration (Nair et al. 2009a, 2009b; Nath et al. 2021); mitigate greenhouse gases (Martinelli et al. 2019); in addition to nutrient input (nitrogen, phosphorus, potassium) into the system by the input of plant residues (Duarte 2011) and introduction of nitrogen-fixing plant species (Marron and Epron 2019). AFS contribute to the balance of fauna and the maintenance of natural enemies of pests and diseases (Buzatto et al. 2018). In addition, AFS reduce direct sunlight incidence on the intercropped species, promote air temperature buffering, minimize moisture loss in the crop microenvironment (Arenas-Corraliza et al. 2022), and protect the soil against leaching and erosion (Rosati et al. 2021).

In addition to multiple ecosystem benefits, AFS have drawn the attention of producers to this niche (Liu et al. 2018; Torres et al. 2016) for providing short-term (as crops in general), medium-term (such as fruits, green biomass, nuts, gums, green manure, oil), and long-term (fiber, pulpwood, timber, fuelwood, and roundwood) products, also attributing higher value to the property (Nair et al. 2021; Wilson and Lovell 2016). From this perspective, timber has the potential to strengthen economic power in the long term as the last harvested product in an AFS.

Tree growth, wood formation, and its resulting quality are affected by numerous factors such as spacing, competition for light, nutrients, and water in the soil (Coelho et al. 2010; Gao et al. 2013). Growing certain tree species in monoculture and especially in AFS is challenging (Liu et al. 2018) because the demand for resources is different for each type of cultivar since they have different physiological needs (Onuwa et al. 2021). For example, crops recquire a greater quantity of nutrients in a shorter time when compared to trees (Schwartz et al. 2021). Therefore, a successful AFS implementation requires synergy between environmental and socioeconomic conditions and correct management techniques and practices (Luedeling et al. 2016). This system also needs more elaborate planning when compared to conventional monoculture, as it requires extensive knowledge of agricultural, livestock, and forestry activities (Luedeling et al. 2016; Viana et al. 2013).

However, generally, most AFS studies regarding its woody component do not analyze its productivity entirely, which creates a gap in individual growth dynamics, competition tolerance, product quality, species choice, and best combination according to particular climatic and environmental characteristics (Liu et al. 2018; Nguyen et al. 2014; Thomas et al. 2021). As wood quality is a predictor for pricing and final product, the influence of growing conditions on the resulting characteristics of wood quality is fundamental to better guide producers into more assertive silvicultural practices (Silva et al. 2018).

Thus, the main objective of this study was to survey articles that report results on the quality of wood produced in agroforestry systems. Moreover, the secondary objective was to identify the gaps in existing knowledge about the quality of wood produced in AFS to direct new studies.

Materials and methods

Literature search and screening

According to the literature, wood quality corresponds to a set of wood characteristics (Vidaurre et al.

[1] [1]The terms "tropical" and "subtropical" is thereafter used considering both in the geographically accurate sense (tropics are considered 23.5° north and south of the Equator; while subtropics are areas located in the middle latitudes from 23°26′10.3″ to approximately 35° north and south of the globe), as well as the definition adopted by Nair et al. (2021): "(…) the word tropics is used in a general sense to include not only countries and regions within the geographical limits of the tropics but also the subtropical developing countries that have agroecological and socioeconomic characteristics and land-use problems that are comparable to those of the countries within the tropical (geographic) limits."

2020) that enable the assessment of its properties by quantitative and qualitative analyses. Based on the results of such evaluations, the most suitable use for the wood as final product is indicated, thus maximizing its potential. Therefore, we decided that the search would report on articles that describes the results of physical, chemical, mechanical, and anatomical analysis and wood properties, as well as its durability and drying process experiments. In addition, to make the search more precise, we included various terms used to designate agroforestry systems, chosen by consulting different works on the subject, including books focused on agroforestry systems (Nair and Garrity 2012; Nair 2014; Nair et al. 2021). Thus, we carried out a manual systematic search on the Scopus (Elsevier™) and Web of Science™ (Clarivate), platforms that allow the use of various keywords and Boolean operators. The search was carried out in November 2023 considering the terms searched in the title, abstract and keywords of the document, limiting the search to article-type documents only. All the searched terms are listed below.

The search was as it follows: (TITLE-ABS-KEY (“agroforestry system” OR “agrosystems” OR “silvipastoral” OR “Consortium” OR “Tree Consortium” OR “agroforestry” OR “Agrosilvopastoral systems” OR “Agrosilvopastoral” OR “Agrisilvicultural systems” OR “mixed systems” OR “agroforestry” OR “agro-forestry” OR “silvopasture” OR “agro-silviculture” OR “agro-sylviculture” OR “intercropping” OR “crop-livestock-forest” OR “integrated crop-livestock-forest”) AND TITLE-ABS-KEY (“timbering” OR “wood pests” OR “wood” OR “log” OR “timber” OR “planks” OR “pallet” OR “wooden box” OR “wooden” OR “tapen” OR “wood-packing” OR “wood packing” OR “wood packaging materials” OR “wood boring” OR “trees” OR “tree”) AND TITLE-ABS-KEY (“wood drying” OR “productivity” OR “chemical composition” OR “physical properties” OR “mechanical properties” OR “wood strength” OR “density” OR “anatomical features” OR “anatomical properties” OR “wood properties” OR “biomass” OR “moisture” OR “anisotropic coefficient” OR “contraction” OR “swelling” OR “density” OR “calorific value” OR “pith eccentricity” OR “heartwood quantification” OR “sapwood quantification OR “chemistry” OR “extractives” OR “ash” OR “lignin” OR “modulus of rupture” OR “modulus of elasticity” OR “shear” OR “Janka hardness” OR “wood anatomy” OR “organoleptic” OR “durability” OR “wood defects” OR “properties of wood” OR “productive sustainability”) AND TITLE-ABS-KEY (“wood qualit*”).

We did not limit the publication year, therefore articles published until 24th November of 2023 were included. During the manual screening process, the inclusion criteria were: (a) the study material must be from an agroforestry system; (b) the study has to analyze some wood quality parameter. We considered analysis of wood quality after tree cutting and before the timber transformation into a final product. All the resulting manuscripts went through a duplicate exclusion process using the *R* studio Software (R Core Team 2018).

During the screening process, the authors manually selected the articles that attended both inclusion criteria simultaneously and analyzed them regarding: published journal, subject area of publication, publication year, keywords choice, and authors. Subsequently, each article was integrally assessed, focusing on their methodology and results, especially on the study area, species and analyzed wood properties. Those topics were then described and discussed, concluding this study with a gap in knowledge regarding the theme and future works indications. The processes of this study are illustrated in the following flowchart (Figure 1).

VOSviewer software (version 1.6.18) was used to build and visualize keyword co-occurrence networks, considering the minimum number of occurrences for each keyword as 1. For this analysis, both the keywords indexed by the platform and the keywords indicated by the authors were considered. The result was plotted as a co-occurrence network.

Results and discussion

Investigative reading and systemic search

The Scopus search resulted in 34 articles, while the Web of Science had 20 results. After the exclusion of duplicates, 35 articles remained. Of these, 22 articles were excluded because they did not focus on wood quality or agroforestry systems themselves. The 13 resulting articles that attended both inclusion criteria simultaneously were published from 2014 to 2023 in nine different journals, with the highest production in 2021.

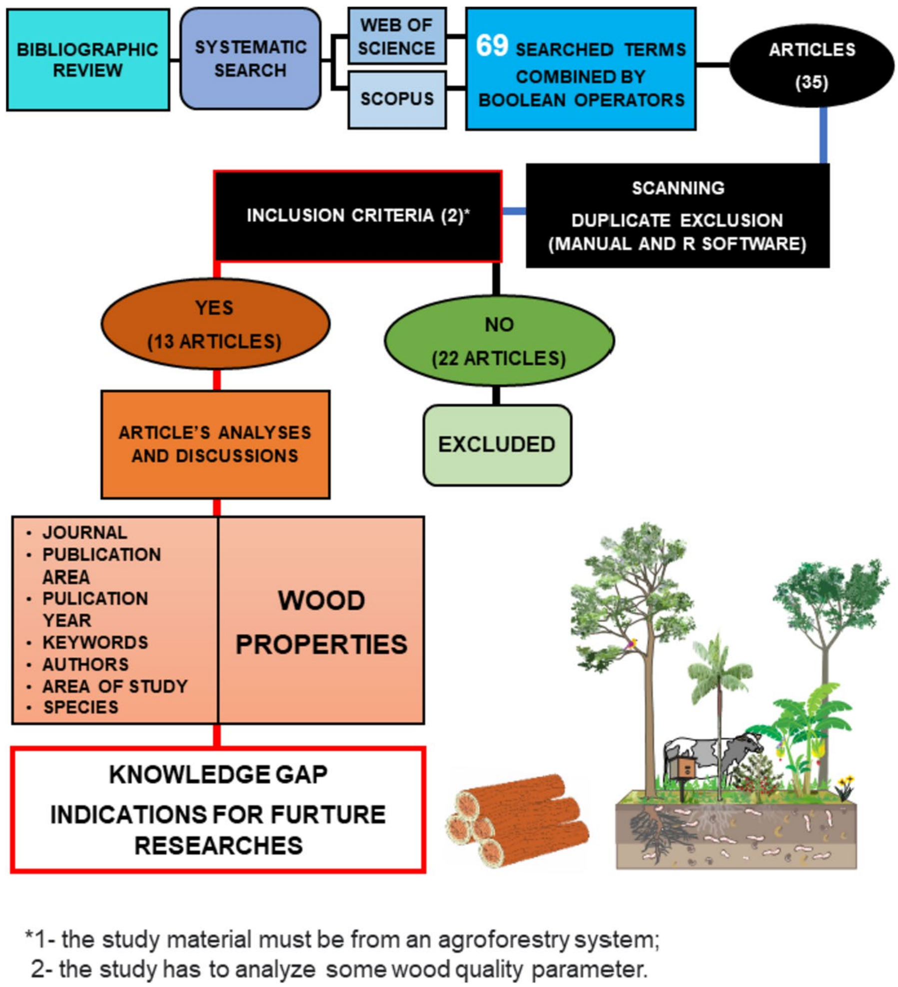

Fig. 1 Flowchart of the search, scanning and selection process

These journals are focused on three subject areas: Agricultural and Biological Sciences (12), Materials Science (3) and Environmental Sciences (1). Analysis of the articles revealed that most authors focus their studies on aspects of wood technology, and are better suited and therefore accepted in journals that specifically cover this area. Thus, this preference probably arises from a better suitability of these studies to the objectives and scope of these journals.

Agroforestry Systems was the main publisher chosen by the authors, publishing four articles, followed by Scientia Florestalis, which contributed two articles. The other publishers (Revista Brasileira de Ciências Agrárias, IForest, Forests, Madera y Bosques, Wood Material Science and Engineering, Wood Research and Revista Internacional de Produtos de Madeira) each contributed an article. Thus, the relevance and interdisciplinarity of these studies are emphasized, promoting significant advances in the field of research in wood technology (in this case specifically the quality of wood from AFS) and related areas.

AFS has been widely studied since the 1970s, marked by the establishment of dedicated institutions such as ICRAF (World Agroforesty Center) in 1977, deemed as the main responsible for organized global efforts to secure AFS practices and derived knowledge as a science-based and integrated approach (Nair and Garrity 2012). However, articles on the quality of wood produced on AFS are recent, beginning in 2014 (1 article), with an increase in publications in 2021 (5 articles) and a second peak at 2023 (3 articles). This recent movement may have been driven by the United Nations Framework Convention on Climate Change and the United Nations Convention to Combat Desertification, which highlighted the relevance of agroforestry as a mitigating tool for climate change impacts (Gonzalez-Roglich et al. 2019; Zhongming and Wei 2019). Despite this increase, there are few scientific articles published on wood quality in agroforestry systems, emphasizing the demand for more information.

For comparison and discussion purposes, a systematic search using only the most widely used term ("agroforestr* system*") for this practice and a second one using various terms for the same practice used on our main search in this study ("agroforestry system" OR "agrosystems" OR "silvipastoral" OR "Consortium" OR "Tree Consortium" OR "agroforestry" OR "Agrosilvopastoral systems" OR "Agrosilvopastoral" OR "Agrisilvicultural systems" OR "mixed systems" OR "agroforestry" OR "agro-forestry" OR "silvopasture" OR "agro-silviculture" OR "agro-sylviculture" OR "intercropping" OR "crop-livestock-forest" OR "integrated crop-livestock-forest") was conducted on the Scopus database. The first search had a total of 4593 scientific articles ranging from 1980 to 2023, with a progressive and significative increase since the '90s until the last reference year (2023), reaching its peak on 2022 (432 articles). In the last decade (2013–2023), there was an increase in publications from 145 to 386 articles, with an expressive increase from 2020 to 2022 (marked by the COVID-19 pandemic). On the second search, a total of 92,068 articles were found, with the first publication on 1921, indicating an almost 60-year gap from the first search, something that is justified by the recent coining of the term "agroforestry systems" as indicated by Nair et al. (2021). In the last decade, the number of publications increased from 3638 to 6920, following the same tendency of expressive increase between 2020 and 2022. The first major increase reported was by the '70s, however, a progressive and mostly steady increase has been observed since the '90s, with some minor slopes in the early 2000's, reaching its peak in 2022 (7305). In the last decade, publications increased from 3638 to 6920. Analyzing the results from these searches and the focus of our study, there is a disparity in articles addressing the topic of AFS and wood quality.

The previous paragraphs show the importance of carefully choosing the correct set and combination of terms on the keywords, title, and abstract for the study purpose, what will significantly impact future searches and if and how they will reach the intended audience. Furthermore, an inadequately choice of terms could also exclude studies fit for one search purpose, underestimating its results. Finally, the correct set of terms can also reduce the scanning and analyzing time during searches since it excludes non-related work, giving the researchers more time to focus on other priorities. Therefore, one must previously study attentively the theme related topics and identify the most relevant and fit terms for a better and higher visibility for the article. All those arguments justify the expressive number of terms (69) used on our study search.

A total of 126 keywords with 126 link strengths was identified and resulted in the keyword co-occurrence net above (Fig. 2). The value of link strength indicates the cohesion of a set of terms used and the extent to which these words are linked to other words, that is, their association through the articles. The terms with the most links were "forestry", "agroforestry system" and "agroforestry", with four occurrences each (i.e. they were present in four articles). Other keywords with the most links were "shrinkage", "wood qualities" and "wood quality", with three occurrences each.

In the keyword network presented by VOSviewer, the distance indicates the strength of the link between two knowledge domains: the greater the distance, the weaker the link between two items, while the size indicates, proportionally, the number of occurrences (Morais and Palha 2021).

Regarding wood quality from AFS, Melo RR is co-author of four articles (30.77%) and Mascarenhas authored three articles (23.08%)—two as first author and one as co-author. Pimenta AS was the third most prominent author, co-authoring three articles (23.08%). There is a juxtaposition of some co-authors in distinct articles, which may indicate a reduced number of researchers in the field and a need for more research groups in this theme.

Most of the studies were conducted in Brazil (a total of seven), mainly in the Amazon biome (five studies), and two in the Atlantic Forest, both from the same research group; two in Central America (both in Costa Rica), with the e same co-author in both; two in Europe (France and Portugal) and two in Asia (India). Literature on tropical regions and Latin countries is preponderant, reflecting the fact that those areas have a higher number of AFS (Nair et al. 2021). Additionally, this also indicates a gap in studies regarding

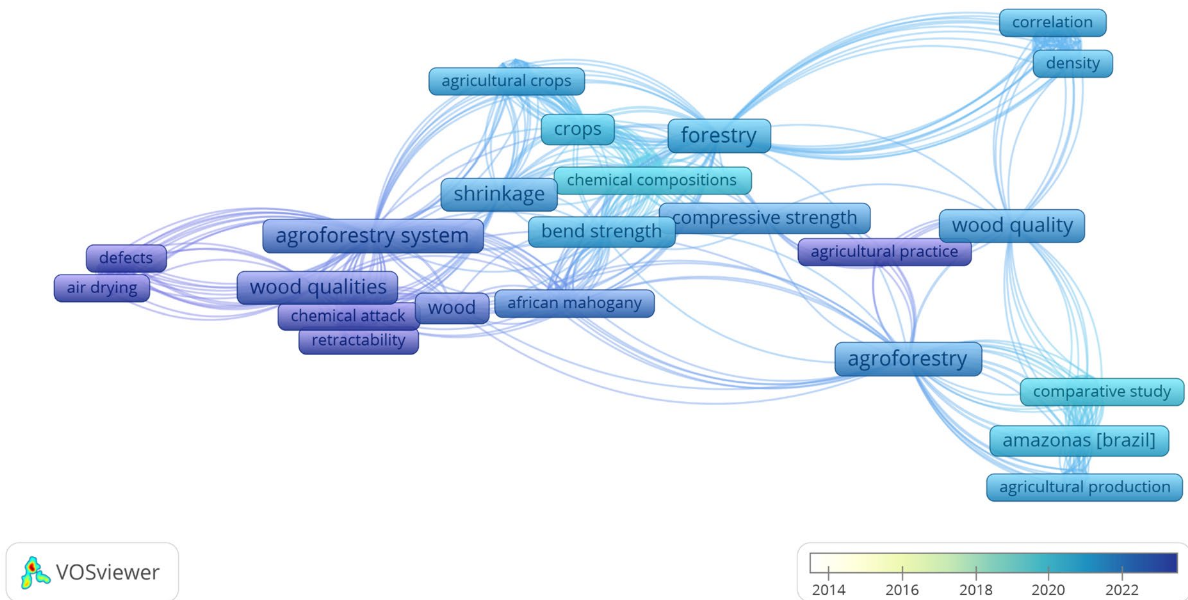

Fig. 2 Keywords co-occurrence net from the analyzed articles

the analysis of the quality of wood produced in AFS in Europe, a region reported to have complex agroforestry systems, and in Africa, a continent that has expanded studies on AFS (Luedeling et al. 2016; Glover et al. 2012).

Each article was analyzed in detail regarding the information on the woody components of agroforestry systems: species, age, wood properties evaluated, and main results of each analysis. This information, as well as the respective references, are detailed in Table 1.

Mangini et al. (2023a, 2023b) carried out studies in Brazil regarding three native species, namely *Parapiptadenia rigida* (Benth.) Brenan, *Peltophorum dubium* (Spreng.) Taub., *Schizolobium parahyba* (Vell.) Blake), and the hybrid *Eucalyptus urophylla* × *Eucalyptus grandis* from an agroforestry system. In the first article (Mangini et al. 2023a), they tested the effect of thermal modification on the dimensional stability under different final temperatures (120 °C, 150 °C, 180 °C, 210 °C) and two periods (2 and 4 hours) by evaluating shrinkage properties and anisotropic index. The thermal treatment significantly modified the dimensional stability of the four studied species and there is a correlation between increasing the treatment temperature and an increase in dimensional stability. The related results are in accordance to literature reports. Therefore, thermal treatment is a great alternative to improve the dimensional stability of wood.

In their second work (Mangini et al. 2023b), they evaluated the drying rate and defects, building a Stepwise based linear regression for drying rate (air dried) versus time. The evaluated defects were: warp (bow and crook), end checks and knots. All the evaluated species had a drying rate in accordance with literature (comparing same genera or botanical family). Regarding the defects, generally, there was an increase in bow warp from primary sawing till the end of the drying process. *Eucalyptus urophylla* × *Eucalyptus grandis* had the highest bow warp percentage (higher than 0, 5%); all the samples had a general average reduction of 47% on crook warp during the drying process, something in accordance with the literature for all the studied species. Regarding the end checks, all the species had an increase of this defect (a general average of 12%), something reported in the literature for the Eucalyptus hybrid, however, there is no data on the literature for comparison regarding the other three species. Three species had knots (*Schizolobium parahyba, Parapiptadenia rigida* and *Eucalyptus urophylla* × *Eucalyptus grandis),* with a total general average percentage of 42%, where *S. parahyba* had the highest average value (75%). All those

Table 1 Summary of key findings on the wood properties from AFS analyzed in each of the reported articles

Article reference and country, *Species* (age i nyears)	Evaluated properties	Main results			
Mangini et al. (2023a) Brazil *Parapiptadenia rigida, Peltophorum dubium, Schizolobium parahyba, Hybrid Eucalyptus urophylla x Eucalyptus grandis* (9)	All the following properties were evaluated for 5 different temperatures (Control; 120 °C; 150 °C; 180 °C; 210 °C)	*Parapiptadenia rigida*	*Peltophorum dubium*	*Schizolobium parahyba*	*Eucalyptus urophylla × Eucalyptus grandis*
	Radial contraction (βRd) at 2 h	5.14; 4.92; 4.37; 4.41; 3.70	3.06; 3.67; 3.15; 3.38; 2.42	3.11; 3.60; 2.95; 3.72; 2.46	4.81; 5.03; 4.90; 4.48; 3.04
	Radial contraction (βRd) at 4 h	5.55; 4.05; 4.08; 4.54; 3.98	3.06; 2.90; 3.07; 3.23; 1.96	3.12; 3.82; 3.09; 2.64; 2.08	4.80; 4.92; 4.66; 4.99; 3.23
	Tangencial contraction (βTg) at 2 h	9.96; 9.24; 9.59; 8.93; 7.38	6.30; 7.61; 6.64; 6.54; 4.45	5.86; 7.02; 5.87; 5.89; 4.38	9.33; 8.94; 9.3; 9.15; 5.63
	Tangencial contraction (βTg) at 4 h	9.95; 9.15; 8.01; 9.78; 9.69	6.29; 5.54; 6.12; 6.52; 3.80	5.88; 6.92; 5.54; 5.34; 3.95	9.33; 8.96; 8.41; 8.87; 6.68
	Longitudinal contraction (βLg) at 2 h	0.92; 0.94; 1.03; 0.83; 0.71	0.94; 0.87; 0.55; 0.68; 0.80	0.96; 0.63; 0.95; 0.89; 0.70	0.62; 0.45; 0.43; 0.47; 0.63
	Longitudinal contraction (βLg) at 4 h	0.91; 1.09; 1.02; 0.81; 0.86	0.74; 0.68; 0.52; 0.63; 0.45	0.99; 0.75; 0.58; 0.44; 0.74	0.69; 0.69; 0.68; 0.52; 0.70
	Anisotropic coefficient of contraction (Caβ) at 2 h	1.94; 1.94; 2.18; 2.13; 2.05	2.13; 2.08; 2.14; 2.01; 1.92	1.96; 1.98; 1.76; 2.06; 1.96	2.00; 1.83; 1.93; 2.08; 1.95
	Anisotropic coefficient of contraction (Caβ) at 4 h	1.92; 2.34; 2.01; 2.19; 1.99	2.14; 1.94; 2.16; 2.07; 2.05	1.95; 1.95; 1.85; 2.13; 1.97	1.98; 1.97; 1.84; 1.80; 2.13
	Radial swelling (αRd) at 2 h	3.10; 3.64; 3.92; 3.71; 2.84	1.66; 2.60; 2.05; 2.18; 1.36	1.50; 2.22; 2.23; 1.71; 1.04	4.37; 4.04; 4.03; 3.69; 2.10
	Radial swelling (αRd) at 4 h	3.13; 3.76; 3.63; 3.44; 3.14	1.65; 1.68; 1.98; 2.10; 0.99	1.41; 1.97; 2.01; 1.70; 0.92	4.11; 3.98; 3.25; 3.61; 2.16
	Tangential swelling (αTg) at 2 h	6.44; 7.52; 8.38; 7.68; 5.54;	3.35; 5.33; 4.25; 4.58; 2.77	3.12; 4.39; 4.65; 3.50; 1.99	8.83; 7.99; 7.75; 7.14; 4.18
	Tangential swelling (αTg) at 4 h	6.96; 8.12; 6.82; 7.48; 3.82	3.35; 3.21; 3.83; 4.37; 1.94	3.18; 3.71; 3.83; 3.29;1.83	8.39; 7.95; 6.29; 7.21; 4.04
	Longitudinal swelling (αLg) at 2 h	0.53; 0.59; 0.56; 0.39; 0.36	0.38; 0.59; 0.58; 0.38; 0.38	0.55; 0.72; 0.55; 0.50; 0.24	0.58; 0.41; 0.23; 0.26; 0.11
	Longitudinal swelling (αLg) at 4 h	0.52; 0.49; 0.63; 0.51; 0.37	0.44; 0.40; 0.39; 0.44; 0.25	0.64; 0.30; 0.34; 0.21; 0.23	0.57; 0.49; 0.37; 0.42; 0.26
	Anisotropic swelling coefficient (Caα) at 2 h	2.27; 2.10; 2.17; 2.11; 1.83	2.13; 2.08; 2.10; 2.16; 2.11	2.20; 2.09; 2.14; 2.11; 2.26	2.10; 2.01; 1.92; 2.00; 2.04
	Anisotropic swelling coefficient (Caα) at 4 h	2.27; 2.18; 1.90; 2.23; 1.88	2.17; 1.94; 2.21; 2.16; 2.31	2.23; 1.94; 2.00; 2.00; 1.94	2.09; 2.09; 1.97; 2.02; 1.96

Table 1 (continued)

	Evaluated properties/species	*Parapiptadenia rigida*	*Peltophorum dubium*	*Schizolobium parahyba*	*Eucalyptus urophylla x Eucalyptus grandis*
Mangini et al. (2023b) Brazil *Parapiptadenia rigida, Peltophorum dubium, Schizolobium parahyba, Hybrid Eucalyptus urophylla x Eucalyptus grandis* (11)	Dry basis moisture content (%)	16.78	25.46	32.27	19.72
	Basic specific mass (g/cm^3)	0.652	0.488	0.277	0.509
	Wood warp: Bow (before drying; after drying)	0.536; 0.788	0.087; 0.012	0.071; 0.133	0.165; 0.604
	Wood warp: Crook (before drying; after drying)	0.428; 0.200	0.037; 0.112	0.154; 0.091	0.026; 0.080
	End checks index for the worst surface (before drying; after drying)	3.328; 3.684	0.681; 2.837	7.358; 6.973	6.307; 7.670
	End checks index for the second surface (before drying; after drying)	2.904; 3.160	0.650; 1.737	5.300; 6.029	5.442; 6.357
	Evaluated properties/species	Intensively managed block (Bim)	Partially managed linear (Lpm)	Unmanaged block (Bum)	Monoculture
Shukla and Viswanath (2023) India *Tectona grandis* (24 and 25)	Equilibrium moisture content (EMC): air-dry (%); green (%)	12.7; 122.1	13.1; 112.3	11.3; 84.3	12.0; 76.0
	Density (g cm^3) and specific gravity (SG)	0.629; 0.55	0.636; 0.57	0.667; 0.60	0.672; 0.604
	Volumetric shrinkages (βr; βt; βv) (%)	2.4; 4.0; 7.4	3.2; 4.7; 8.4	2.8; 4.8; 6.8	2.3; 4.8; 6.8
	Modulus of rupture (MOR) (MPa)	87.64	84.97	98.12	94.08
	Modulus of elasticity (MOE) (GPa)	9.26	8.75	11.54	11.73
	Fiber stress at limit of proportionality (FS at LP) (MPa)	52.44	49.82	59.27	63.90
	Maximum crushing stress (MCS) (MPa)	41.59	44.65	51.14	52.19
	Compressive strength at compression of 2.5 mm (MPa)	14.91	13.08	12.53	–
	Compressive strength perpendicular to grain (MPa)	8.36	9.62	9.28	9.91
	Surface hardness (kN)	4.35	3.65	4.53	5.14
	Side	4.45	4.45	4.11	4.79
	End				

Table 1 (continued)

	Evaluated properties	Site 1	Site 2		
Sousa et al. (2021) Portugal *Quercus rotundifolia* (ni)	Ring width (RW) (mm)	1.81	1.55		
	Ring density (RD) (g cm^{-3})	0.914	0.888		
	Minimum (MND) and maximum ring density (MXD) (g cm^{-3})	0.840; 0.991	0.820; 0.951		
	Evaluated properties	*Castanea sativa* Mill	*Quercus robur*	Hybrid *Juglans nigra x regia*	Hybrid *Populus deltoides x Populus tristris*
Terrasse et al. (2021) France *Castanea sativa* (9, 8 and 6) *Quercus robur* (18, 15 and 11) Hybrid *Juglans nigra x Juglans regia* (14, 13, and 12) Hybrid *Populus deltoides x Populus tristris* (14, 12 and 10)	Wood density (g cm^{-3})	0.76	0.68	0.59	0.48
	Extractive content	3.2	7.9	5.4	2.0
	Durability class	5.0	4.0	5.0	5.0
Mascarenhas et al. (2021a) Brazil *Schizolobium parahyba* (19)	Basic density: 0.354 g cm^{-3}			Compression-parallel-to-grain strength: 22.66 MPa; Parallel-to-grain stiffness: 7786 MPa; Perpendicular-to-grain strength: 4.67; Perpendicular-to-grain stiffness: 1433 MPa; Parallel-to-grain: 1520 N	
	Apparent density: 0.280 g cm^{-3}				
	Volumetric shrinkage: 8.03%				
	Anisotropic coefficient: 1.93			Static bending stiffness: 4992 MPa; Static bending strength: 34.57 MPa	
	Chemical components: (extractive content: 4.03%; lignin content: 29.1%; ash content: 0.91%)			Janka hardness—perpendicular-to-grain: 1340 N	
	Holocellulose content: 65.96%			Shear strength: 3.62 MPa	
Mascarenhas et al. (2021b) Brazil *Khaya senegalensis* (19)	Basic density: 0.52 g cm^{-3}			Parallel-to-grain compression strength: 26.5 MPa	
	Anisotropic coefficient: < 1.3%			Static bending strength: 56.0 MPa	
	Chemical components: (extractive content: 6.17%; lignin content: 25.80%; ash content: 0.8%; holocellulose content: 67.23%)			Janka hardness parallel-to-grain: 3420.0 N	

Table 1 (continued)

			Shear strength: 10.4 MPa	
	Evaluated properties/samples	9 years old tree-agroforestry	10 years old tree-agroforestry	10 years old tree-plantation
Segura-Elizondo and Moya (2021) Costa Rica *Cedrela odorata* (9 and 10)	Dasometric properties (%) Bark Sapwood Heartwood	13.85 70.75 14.90	12.95 68.57 17.97	15.00 72.96 11.47
	Specific gravity Apparent density (g cm^{-3})	0.32 0.58	0.31 0.57	0.29 0.60
	Green moisture content (%)	82.23	83.09	108.15
	Volumetric shrinkage (%)	8.97	8.95	9.54
	Janka hardness (N) Lateral Axial	1418.3 1704.5	1458.0 2002.2	1283.0 1485.1
	Compression stress (MPa)	16.35	17.19	16.82
	Tension stress (MPa)	41.30	40.46	49.44
	Flexion: MOR (MPa); MOE(GPa)	28.98 4.07	30.23 4.46	24.56 3.70
	Shear strength parallel-to-grain: tangential stress; radial stress	6.78 6.10	6.77 6.36	6.65 6.34
	Chemical components (%): extractive content (ethanol–toluene); lignin content; ash content; cellulose content	7.33; 30.89; 1.81; 53.00	12; 30.72; 0.6; 52.50	11.33; 28.64; 0.68; 51.83
	Net calorific power (kJ/kg)	17189.6	18471.1	19138.9
	Content (%): carbon; hydrogen; nitrogen	45.58; 6.34; 0.15	45.65; 6.35; 0.13	45.69; 6.36; 0.18
		Evaluated properties/type of plantation	Monoculture	Agroforestry system
Ferreira et al. (2020a) Brazil Hybrid *Eucalyptus grandis x Eucalyptus urophylla* (5)		Basic density (g cm^{-3})	0.403	0.433
		Growth stress (longitudinal residual strain)	0.114	0.147

Table 1 (continued)

	Evaluated properties/type of plantation	Monoculture	Agroforestry system
Ferreira et al. (2020b) Brazil Hybrid *Eucalyptus grandis x Eucalyptus urophylla* (7)	Density (g cm^{-3})	0.546	0.574
	Volumetric shrinkages (βr; βt; βv) (%)	3.57; 5.99; 9.53	3.89; 6.14; 9.96
	Anisotropic factor (Af) (%)	1.59	1.63
	Modulus of elasticity and rupture obtained in the static bending (MPa)	7,396 71.20	9,454 78.13
	Modulus of elasticity and rupture obtained in the parallel-to-grain compressive (MPa)	12,634 26.90	13,901 40.29
	Compressive strength (MPa)	8.61	8.68
	Janka hardnes (MPa)	3,482	3,663
	Chemical components (%): ash content, extractives content, lignin content and total carbohydrates	0.43; 2.40; 23.95; 73.22	0.47; 3.23; 22.44; 73.86
	Evaluated properties/type of plantation	Monoculture	Agroforestry system
Silva et al. (2019) Brazil *Schizolobium parahyba* (3)	Basic density (g cm^{-3})	0.264	0.246
	Apparent density (g cm^{-3})	0.518	0.488
	Compressive strength (MPa)	25.37	23.06
	Modulus of elasticity (MPa)	2048.1	1824.8
	Modulus of rupture (MPa)	30.36	34.41
	Modulus of elasticity (MPa)	5031.7	5016.8

Table 1 (continued)

	Evaluated properties/population	Population 1	Population 2
Rigg-Aguilar and Moya (2018) Costa Rica *Cedrela odorata* (7)	Diameter at breast height (cm)	13.9	19.7
	Bark (%)	21.3	12.8
	Pith (%)	0.6	0.3
	Pith eccentricity (%)	5.7	4.0
	Heartwood (%)	24.5	31.2
	Sapwood (%)	53.5	60.2
	Volumetric shrinkage (%)	9.29	6.82
	Green density (g cm^{-3})	0.62	0.50
	Specific gravity (g cm^{-3})	0.27	0.26
	EEM in bending (MPa)	4043	5769
	MOR in bending (MPa)	27.33	44.14
	Compression stress in the longitudinal direction (MPa)	12.72	23.55
	Janka hardness in the axial direction (N)	72.58	78.46
	Janka hardness in the lateral direction (N)	14.24	15.46
	Tension stress parallel to grain (MPa)	41.70	44.09
	Tension stress perpendicular to the grain (MPa)	1.98	3.23
	Shear stress parallel to grain (MPa)	4.93	4.59

Table 1 (continued)

	Evaluated properties/type of managment	Intensively managed block (Bim)	Partialliy managed linear (Lpm)	Unmanaged block (Bum)
Shukla and Viswanath (2014) India *Tectona grandis* (12)	Specific gravity	0.559	0.572	0.606
	Volumetric shrinkage (%)	9.14	10.11	8.05
	Modulus of rupture-MOR (MPa)	82.10	74.48	84.34
	Modulus of elasticity-MOE (GPa)	8.19	6.96	10.10
	Fiber stress at the limit of proportionality—FS at LP (MPa)	54.99	42.95	54.59
	Compressive strength parallel to grain (MPa): max. crushing stress (MCS)	42.13	38.20	40.94
	Compressive strength perpendicular to the grain (MPa)	9.71	9.53	10.78
	Surface hardness (kN)	3.74	3.99	5.08
	Side	4.09	3.75	4.55
	End			

Some studies analyzed the same property but with different denominations and we chose to maintain the same terminology as their respective article. ni: not informed. In cells with more than one value, each value corresponds to a variation from the analyzed parameter with their respective order established on the evaluated parameter information cell defects lower the wood quality, reducing its value and limiting its use.

Shukla and Viswanath (2023) evaluated and compared wood quality parameters (equilibrium moisture content, density, shrinkage, static bending strength and stiffness, compressive strength parallel and perpendicular to grain, hardness, nail and screw holding powers) of *Tectona grandis* L.f. plantations grown under three agroforestry practices (AFPs): intensively managed block (Bim, with a 4 × 4 m spacing and 25 years), unmanaged block (Bum, with a spacing of 2 × 2 m and 24 years), and partially managed linear (Lpm, with a 4 × 4 m spacing and 25 years), in India. Those authors compared them with what they labeled as 'forest teak', i.e. mature teak from 14 different localities. Bum had better results for the evaluated properties when comparing the three AFPs, indicating that the management (or the lack of it in this case) has influence on the wood properties (Table 1). When comparing the AFPs with the 'teak forest', the latter had better results, which indicates that age (since the 'teak forest' are denominated as mature) has also some significant influence on the teak wood quality.

Sousa et al. (2021) aimed to study wood density and variation in tree ring width of *Quercus rotundifolia* in two locations in southern Portugal, a drought-sensitive region to climate change and land desertification. The wood from both sites had high densities, with slightly different values (Table 1), and this property has a positive and significant relationship with ring width. The authors also mention that these discoveries contribute to the use of this species for high-value wood products.

The following study analyzed the properties of branches from four species—*Castanea sativa* Mill (9, 8, and 6 years), *Quercus robur* (18, 15, and 11 years); hybrid *Juglans nigra* × *Juglans regia*—also referred to as *Juglans* × *intermedia* (14, 13 and 12 years); hybrid *Populus deltoides* × *Populus tristris*, also referred to as *Populus* × *generosa* (14, 12 and 10 years)—from two French regions (Terrasse et al. 2021). This study aimed to stimulate the use of these four species and increase the profitability for AFS farmers. Additionally, the authors reported that the branches of the studied species have the potential for the manufacture of biomaterials based on their density—medium to high (Table 1). Lastly, Terrasse et al. (2021) do not indicate the external use of the aforementioned species, given their low extractives content and, in turn, lower natural resistance to rotting.

Mascarenhas et al. (2021a) studied the technological characteristics of *Schizolobium parahyba* var. *amazonicum* (paricá) wood from an agroforestry system in Rondônia (Brazil), indicating its industrial applications. These authors found that the studied wood (Table 1) had similar characteristics to wood from monoculture and natural forests, and specific strength similar to species commonly used in Brazilian construction. Thus, *Schizolobium parahyba* var. *amazonicum* wood was recommended for engineered panels and high-value-added products. Finally, the authors highlighted agroforestry systems as a viable option for good-quality paricá wood production.

A study by Mascarenhas et al. (2021b) discussing the same aforementioned parameters (previous paragraph) (Table 1) was conducted with African mahogany (*Khaya senegalensis*) from an AFS in Rondônia (Brazil). The African mahogany wood had suitable characteristics for light and structural uses due to its static flexural strength, grain parallel compression, and hardness. The authors concluded that the chemical composition and mechanical properties of African mahogany wood produced in AFS are very similar to the properties of the *Khaya* genus grown in natural forests.

Segura-Elizondo and Moya (2021) compared the quality of *Cedrela odorata* wood from AFS (with nine and ten years) with that from monoculture (10 years), both located in Costa Rica (Table 1). The study analyzed the influence of different kinds of planting on wood quality. The authors reported that the agroforest system positively influenced the quality of wood since agroforest systems provide better conditions (such as more fertile soils) for tree development (Marron and Epron 2019; Duarte 2011). In this study, trees from AFS had larger diameters, a higher heartwood percentage, a higher modulus of rupture and elasticity, and a higher percentage of extractives (ethanol–toluene) compared to monoculture.

Ferreira et al. (2020a, b) studied the wood quality of the *Eucalyptus urophylla* × *Eucalyptus grandis* hybrid, using seven year-old trees from conventional homogeneous plantation and integrated crop-livestock-forest (iCLF) systems, both located in Sinop, Mato Grosso (Brazil). For this purpose, their chemical composition and physical–mechanical properties were analyzed. Wood from the iCLF system had higher

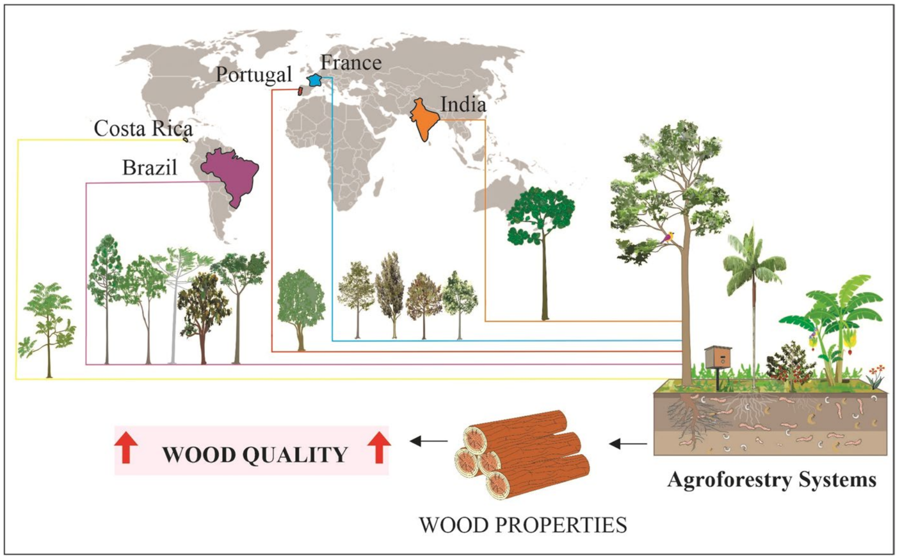

Fig. 3 Graphical summary of origin country, species and the quality of wood from AFS

density (partially explained by the higher growing space between plants), higher contents of ash, extractives and holocellulose when compared to the homogeneous plantation. Additionally, the wood from iCLF also had higher mechanical strength, that is, higher values of static bending and compressive strength. On the other hand, linear and volumetric shrinkage as well as the anisotropic factor were not influenced by the management system. Finally, the authors affirm that the wood from iCLF can be employed for industrial uses, especially sawn lumber and firewood.

The growth, production, and wood quality of the *Eucalyptus grandis x Eucalyptus urophylla* hybrid, cultivated in AFS and monoculture, both located in Sinop, Mato Grosso (Brazil) was evaluated by Ferreira et al. (2020a). The authors reported that the wood from AFS had higher densities, however, it also had the highest longitudinal residual deformations (Table 1). These characteristics were caused by growth stress since the AFS was arranged in blocks with three rows of trees 15 meters apart, whereas the monoculture was arranged in a single block. According to the authors, wind exposure may have caused reaction wood and a possible increase in density and stress.

The following study by Silva et al. (2019) reports the influence of planting type (monoculture and AFS), on the characteristics of *Schizolobium parahyba* var. *amazonicum* (*paricá*) juvenile wood from a farm in Pará (Brazil) (Table 1). In this work, the AFS nutrient input, supported by soil chemical analysis, contributed to higher tree growth. However, the high presence of phosphorus in the soil—a nutrient that may limit cambial activity—contributed to the wood's lower basic density of the wood (0.246 g cm^{-3} in AFS and 0.264 g cm^{-3} in monoculture). Regarding the wood anatomy, there were vessels with smaller diameters (158.85 μm in AFS and 170.66 μmm in monoculture); lower ray frequency (5.55 mm^{-1} AFS and 6.43 mm^{-1} monoculture); and lower mechanical resistance to parallel compression (23.06 MPa in AFS and 25.37 MPa in monoculture). The other anatomical characteristics (fiber length and diameter; lumen and wall thickness; vessel frequency and length; ray height and width) of wood from AFS showed no statistically significant differences compared to monoculture.

Two 7-year-old *Cedrela odorata* populations of the same provenance, both at 3x3 meter spacing in the same locality in Costa Rica and with the same species combination (*Theobroma cacao*), were analyzed to identify differences between morphological characteristics, physical and mechanical properties of the wood (Rigg-Aguilar and Moya 2018). The results showed differences between morphological characteristics (DAP, percentage of bark and pith); Population 1 had higher values for physical properties (volumetric shrinkage and moisture content); while Population 2 had higher values for mechanical properties (MOR and MOE, longitudinal compression and perpendicular tension). This study highlighted the importance of studying and understanding the genetic variation of seminal seedlings and their influence on wood properties—indispensable knowledge for an adequate destination and correct use. Thus, Population 2 presented growth and quality characteristics fundamental for the structural and furniture sector.

Finally, we report a comparative study by Shukla and Viswanath (2014) evaluating the growth, quality, and financial return of teak (*Tectona grandis*) wood planted in Karnataka (India) in agroforests with three different management practices (unmanaged 2 m row planting; unmanaged block; and intensive management with 2×2 m^2 spacing—all with 12 years). The wood quality parameters, such as density, modulus of elasticity, compression parallel to the grain, and hardness of the unmanaged block plantation were statistically superior to the others (Table 1). However, the wood from the three types of management had lower quality when compared to monoculture ones—something expected since the trees from AFS were younger. Despite this, the authors state that unmanaged blocks may be a good choice for commercial exploitation of the species since the lower cost of plantation management offsets the resulting wood quality.

The works indicate the properties and potential use of wood from AFS, as well as the employment of branches of several species to manufacture biomaterials; some properties of *Schizolobium parahyba* var. *amazonicum* wood are similar, while *Khaya senegalensis* wood is very similar to individuals from natural forests (Figure 3). Moreover, more fertile soils are also reported for AFS (because of the higher organic matter deposition) when compared to monoculture, along with higher volumetry and quality of *Cedrela odorata* wood grown in AFS. Furthermore, limiting factors are also reported, such as increased phosphorus content and its contribution to the limitation of cambial activity; wide spacings are reported to have relation to higher longitudinal residual deformations in *Eucalyptus* wood.

Another reported aspect is the drying process of wood from AFS where the studied species showed similar patterns of rate drying and defects as reported in the literature regarding material from other types of management. Furthermore, wood from AFS is also reported to have similar dimensional stability after thermal modification as those reported in the literature for material from different sources and managements. Regarding the properties of teak, the studies here reported indicate that unmanaged systems have a positive correlation to desirable wood qualities values, with wood from AFS having a good evaluation considering their age when compared to other sources.

Overall, in the reported articles, the species grown in AFS are of exotic/introduced origin, with some native to the countries where they are planted. In France, the species used were *Castanea sativa,* a native species exploited for chestnut and wood production (INE 2017); *Quercus robur,* also native, used for barrels production, construction, and furniture (Blanc-Jolivet and Liesebach 2015); the natural hybrid *Juglans nigra x regia (Juglans x intermedia*) is used for rootstock, timber and nut production (Woeste and Michler 2011; Juravle 2020); and the introduced hybrid *Populus deltoides x Populus tristris*, is used for energy and timber (Sannigrahi et al. 2010). In Portugal, *Quercus rotundifolia* Lam. is characteristic of the Portuguese 'montado', a Mediterranean typical agrosilvopastoral system, and its economic importance is mainly related to acorn production for animal grazing and human consumption, and its wood is used for charcoal production and firewood (De Rigo and Caudullo 2016; Sousa et al. 2021).

In Brazil, the exotic species *Khaya senegalensis* is used for furniture (Pinheiro et al. 2011), and the also exotic hybrid *Eucalyptus grandis x Eucalyptus urophylla* is employed for pulp and paper production, charcoal for steelmaking, wood panels, and flooring (IBÁ 2020). Both species are mainly cultivated to attend the international market. *Schizolobium parahyba*, native to Brazil and the object of two articles in this review, is widely used in lamination industries for panels and boxes, in addition, such employment stimulates the use of this species in forestry and agroforestry scenarios (Cordeiro et al. 2020). *Parapiptadenia rigida* (Benth.) Brenan and *Peltophorum dubium* (Spreng.) Taub are also native species that are used for firewood and charcoal, as well as for construction, being the last species commonly used in urban afforestation (Carvalho 2003a, b).

Cedrela odorata, a native species to tropical regions of the Americas, was the object of two studies in Costa Rica and it is highly appreciated for its wood of excellent quality for furniture, plywood, and veneer (Lopez et al. 2020). In India, the studied species was

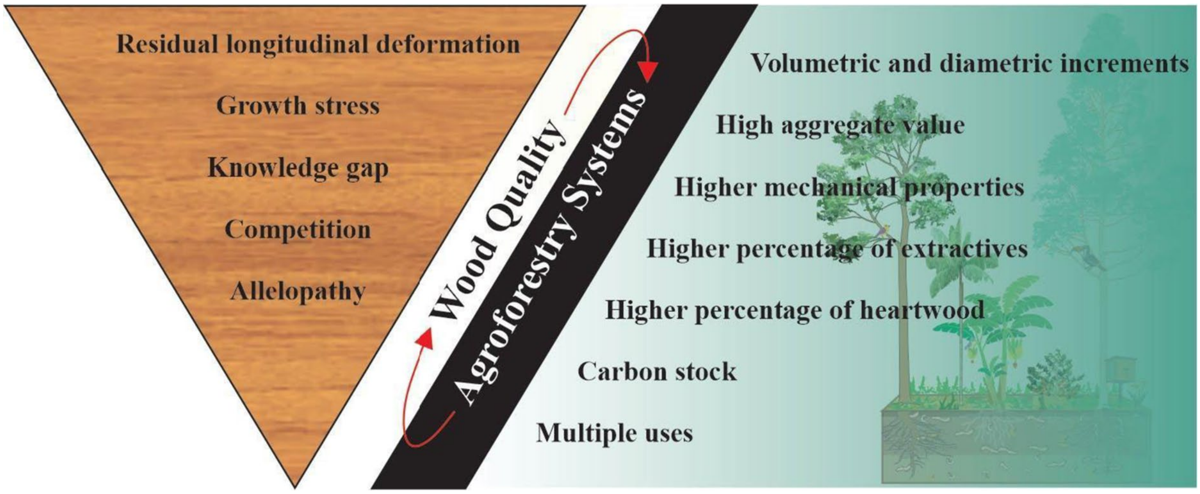

Fig. 4 Quality of wood grown in AFS when compared to monoculture, with their respective benefits and challenges

Tectona grandis, a native species that is part of the market for high-value-added wood with multiple uses, including timber, pulp, energy, and non-timber products (Chávez-Salgado et al. 2020).

Except for *Parapiptadenia rigida*, *Peltophorum dubium* and *Quercus rotundifolia* (species that are usually implemented for restoration, conservation purposes and AFs), all the reported species in this work are already commonly planted in monocultures and also have an established commercial value both in the domestic and international market. However, it is still necessary to produce information about tree species in AFS, as well as to study the different effects and products from these interactions to diversify the AFS composition. For producers interested in diversifying the design of their system, there are already some studies listing forest species of multiple uses with potential employment in these systems (Albuquerque and Silva 2008; Nair et al. 2021; Sales and Araújo 2004).

General perspectives of wood production in AFS

The expansion of AFS on agricultural land is a potential alternative for carbon sequestration and storage (Nath et al. 2021; Rigueiro-Rodríguez et al. 2009). Even though, for this purpose, the maintenance of the tree in the system is of greater relevance, its conversion to wood products is still a valuable benefit, especially as a replacement for fossil and non-renewable products (Geng et al. 2017; Liao and Ning 2023; Oluwadare 2011). Additionally, carbon storage in long-life wood products (like plywood and lumber) represents about 50% of the carbon present in the stem (Sharma et al. 2016). For this trade-off to be worth it, that is, to justify the removal of the woody component and therefore a valuable part of the system's carbon storage, the rotation time should be longer alongside the adoption of slow growing species, as well as the final product should be preferable of long-life in employments such as buildings and furniture (Báder et al. 2023).

Overall, all the listed works report important and unprecedented information on various properties of wood from AFS. Some studies exclude the site as a variable since they considered the monoculture of the tree species alongside the AFS at the same site: those studies analyze a single variable (monoculture × AFS) that would influence the wood quality since the other interference factors (age, progeny, spacing, etc.) were homogenized. Other studies were restricted to comparisons with monoculture in other locations, of other progenies, different spacing and ages, or comparisons with individuals from natural forests.

Some studies report that the characteristics of the wood species used (Figure 4) from AFS are equivalent or superior to those of monoculture. Therefore, this reinforces the notion that trees grown in AFS have attributes compatible with both high value-added uses such as engineered panels and light structural items, which could increase the farmers' profit.

However, only one resulting study of our search reports on the influence of different types of management (or the lack of it, namely Bim, Bum and Lpm from Shukla and Viswanath 2014, 2023) on the quality of wood from AFS, thus this type of questioning appears to be a gap on this specific type of study. This gap is important because numerous external aspects, especially site and management related, have influence (direct and indirectly) on the tree development and consequently, on its wood. Before implementing an agroforestry system, an extensive search is advised to achieve a more efficient use of the available resources and the potential of the adopted species. A solid planning could also mitigate or sometimes even avoid entirely the consequences of unforeseen events—such as pests and diseases, market fluctuations, and weather extremes caused by climate change, especially in later developmental stages of the system (Morhart et al. 2015).

While planning, the farmer must consider the site characteristics such as type of soil (especially nutrient content and soil aeration); climate variables, especially temperature, precipitation (amount and annual distribution); water regime (especially holding capacity and stagnancy of soil moisture). One must also consider the space availability on the land, how much of it the system can occupy, and their topographical conditions. From this point, the farmer must see which species can fit and prosper under those conditions, giving preference for native species, and therefore planning to expect the best outcome for the final product or employment.

It is possible to make some generalizations regarding AFS to wood production (such as made by Mohart et al. 2015), however there is a lot of specific parameters to take into account that cannot be generalized (nor it is recommended to be so), since there is a significant number of variables and an even greater possible number of combinations regarding the site alone. Moreover, numerous factors can affect tree development and they must be accounted for when planning to better attend the plant physiological demands, that also vary according to the species and age. Finally, the final product will also impact the choice of management strategy and techniques. Thus, a previous and meticulous study is indispensable and should be highly focused for each specific condition, such as Patel et al. (2017) and Yakob et al. (2014).

By testing the numerous possibilities for arranging the various factors related to AFS (choice of tree species, design, silvicultural treatments, and management), scientific research becomes an essential tool to better align farmers' resources by pointing to the appropriate and most cost-effective management strategy on a commercial scale. Therefore, it is of utmost importance the documentation (description of the processes through technical notes, manuals, or scientific articles) and dissemination of knowledge in an accessible and easily understandable way for those interested in this practice.

The authors Oluwadare (2011) and Sollen-Norrlin et al. (2020) further add that it is essential to introduce financial incentive policies, such as subsidies and payments for ecosystem services provided by AFS. In addition, appropriate marketing strategies are needed to raise awareness about the various products generated by AFS for consumers. Finally, landowners need more training opportunities on how to run a diversified and economically viable AFS.

Although AFS are widely spread on a global scale under different approaches (Nair et al. 2021), wood quality from AFS is a topic still little explored. Crops such as eucalyptus (Pezzopane et al. 2021), acacia, cedar, and mahogany (Akter et al. 2022) have thousands of hectares under intercropping mainly in tropical countries, however, information is scarce on the influence of agroforest systems on the quality of the wood produced. The absence of this type of information generates doubts and highlights the demand for information on how those systems can provide socioeconomic and environmental stability to the sector while maximizing wood quality for use in the final product chain.

In general, there is little information on the quality of wood from agroforestry systems, a gap to be filled by future research. The choice of non-fitting and proper keywords by the authors could influenced our search results since the search engines partially relies on them. For future researches, we suggest the use of terms such as "chemistry" OR "mechanical analysis" AND "agroforestr* system*", as well as related terms of each theme/category, for example. The proper choice of keywords will also facilitate the employment of search engines and databases such as Scopus and WoS, main platforms, alongside Science

Direct and the South American SciELO, that gather data from scientific publications worldwide. The indexation of these data in such platforms is based on several criteria, thus, the data that does not attend all the required parameters are left out during systematic research. Another limitation is that some platforms limit the number of searchable terms as well as Boolean operators.

Conclusions

AFS have numerous macro and local environmental and economic advantages, and this system is especially compelling for small farmers since there is an economic return through different timespans while also providing ecosystem services in such a diverse cropping system. However, the main limiting factor for the wide use of agroforestry systems (AFS) is the need for standardized procedures for their design and management, as well as their carbon stock potential.

There is a difficulty in planning designs that homogenize silvicultural factors—such as spacing, age, pruning, thinning, irrigation, and fertilization—and at the same time isolate the plantation type (monoculture vs. AFS) as the only factor that will influence the results, which makes comparisons superficial.

Studies comprising prolonged periods of implementation and management of AFS are needed for better monitoring of wood quality, with more detailed descriptions of all the system components, their interactions, and resulting products. In addition, more work involving visual, physical, chemical, and mechanical characterization, as well as market preferences and potential uses are needed for a better valuation and market acceptance.

The main gaps found were the reduced number of articles, which reflects a still restricted number of researchers on wood quality produced in AFS; a limited geographic distribution, where the studies focused on only five countries (Brazil, France, Costa Rica, India and Portugal) with no studies on African or Oceanian countries. Moreover, we identified a reduced number of species studied compared to the diversity of the established ones already in use in AFS. Finally, there is still a gap regarding species as *Pinus* spp., *Eucalyptus* spp., *Tectona grandis*, *Khaya* spp., *Acacia* spp., among others, whose wood is already widely studied for its commercial value, but with no information about its use in AFS, as well as the impact of AFS on the quality of its wood.

In order to optimize the results of research related to this topic, a careful approach is recommended when using scientific platforms. The ability to carry out precise and strategic searches is crucial to proper extract relevant information and by doing so, also improves the effectiveness of the scientific research. The combination of search terms used in this work can help speed up the search process, as it combines several terms used to refer to AFS as well as several terms commonly used in wood quality studies, serving as a basis for future searches, in which future researchers can add more terms in accordance with their objectives. In addition, the database obtained in this study and its analysis also optimizes future searches on the subject by gathering all the publications in a single document.

For future studies, we indicate new studies exploring political and legislative contextualization regarding the incentive and regularization tools for AFS (both local and international), especially with a focus on wood production. Another indication for future studies concerns the investigation of the existing lists of underexploited multipurpose woody species with potential for implementation in AFS, and elaborating a comparison according to each biome.

In the articles found as a result of our systematic search, no information was identified on how climate change can affect the characteristics of wood from AFS, nor on strategies to overcome such changes in order to ensure the production of quality wood in these systems. In addition, there is a lack of information on the development of genotypes of woody species that are resistant to unfavorable conditions. There are also gaps in how current AFS models can be designed to optimize wood production for various purposes. Filling these gaps is an opportunity to improve design strategies and management of AFS with a focus on wood production, which could even lead to a new market opportunity in the timber industry.

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Declarations

Conflict of interest All authors certify that they have no affiliations with or involvement in any organization or entity with any financial interest or non-financial interest in the subject matter or materials discussed in this manuscript.

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